

STAT 490S FINAL PROJECT

Introduction

New York City has one of the busiest night scenes in the world. As expected, the city receives thousands of noise complaints per year. Noise complaints are received by the NYPD's hotline for non-emergencies, either through calls (311) or their online website. Grievances filed are expected to be specific, with exact addresses, types of noise it is, length of duration, and the date/ time. This analysis focuses on a subset of the *2016 Parties In New York* data from Kaggle. Credit to Evgenii Vasilev on creating this by sourcing this from the NYC open data portal. The main CSV contains over 225,000 noise complaints from 2016 in New York City. However, a secondary CSV contains every location tagged as *club/bar/restaurant* with at least 10 noise complaints in the last 5 years (2012 - 2016). Every single location has information about the borough and city / neighborhood it's in, specific coordinates and the number of calls. From now on, I will refer to all of these entertainment spaces as clubs for simplicity. For the purposes of statistical analysis, the data will be assumed to be independent and identically distributed. There will be no assumption of normality, because the majority of people go to a few nightclubs and the extrapolation will be that a few nightclubs will dominate noise complaints. The distribution is clearly skewed right. ([Figure 1](#)). Because around 74% of clubs in NY are shared between Brooklyn and Manhattan, they are obviously the loudest boroughs. However, the question we'll try to answer is which Borough has the average loudest entertainment spaces?

Population Parameter

Because the data omits entertainment spaces with less than 10 noise complaints, this is not the entire population. The true population parameter cannot be known, thus affecting all statistical methods below and in the rest of the paper. I will assume for this section that the mean of the dataset is the true population parameter. Defined as the average amount of noise complaints received in the last 5 years. The formula is $\bar{X} = 1/n \sum x_i$. The parameter is unbiased, as by the Law of Large numbers the sample mean converges to the true population mean. The variance is defined as sigma squared divided by n. And because the decomposition of the MSE is variance plus bias squared, the MSE and the variance are equal because the bias is zero. Using the group by function in R, I split the data into each respective borough, where a simulation was run for each area. By choosing a sample size of 50, the CLT general baseline of 30 is satisfied. Despite the rightward skew, the sampling distribution is approximately normal. Repeatedly sampling with replacement 100 times, we get a mean estimator for each borough. Manhattan was 37.3, Bronx was 26.47, Queens was 38.94, Brooklyn was 38.17 and Staten Island was 35.8. I then compared the estimators to the original "[mean](#)". In each respective order, the bias was 0.2, 0.47, -0.26, 0.37 and 0.6. The variance was significant ranging between 12 and a staggering 113. This means the sample means are volatile and suggests the existence of outliers. The final MSE calculations are Bronx (13.15), Staten Island (17.84), Queens (55.62), Manhattan (67.76) and Brooklyn (113.98). This suggests that the sample mean is a more efficient estimator for Bronx and Staten Island compared to Manhattan and Brooklyn. I would hypothesize that this is because there is more variability in the noise complaints inside of these areas.

Confidence Intervals

Wald-type confidence intervals are the best fit for this scenario. Because $X_1, \dots, X_n$ are assumed to be iid and the distribution is not normally distributed, we can use the Central Limit and Slutsky's theorems to construct an asymptotic interval of where the true population parameter lies. As n approaches infinity, $\bar{x} \pm z_{1-\alpha/2} * (\sigma / \sqrt{n})$ becomes very accurate. Picking the alpha to be 95%, the $qnorm(0.975)$ becomes 1.96 for the critical value. Using n as the amount of clubs in each borough and the sample mean and the standard deviation of each area, the calculations are as follows. ([Figure 3](#)). The interpretation is that upon repeated sampling of data, 95% of intervals constructed in the same way would contain the true average number of noise complaints for a given bar in a Borough. There is no strong indicator in this analysis of any one borough being significantly above the rest, with Queens, Bronx and Manhattan all being in the same general range.

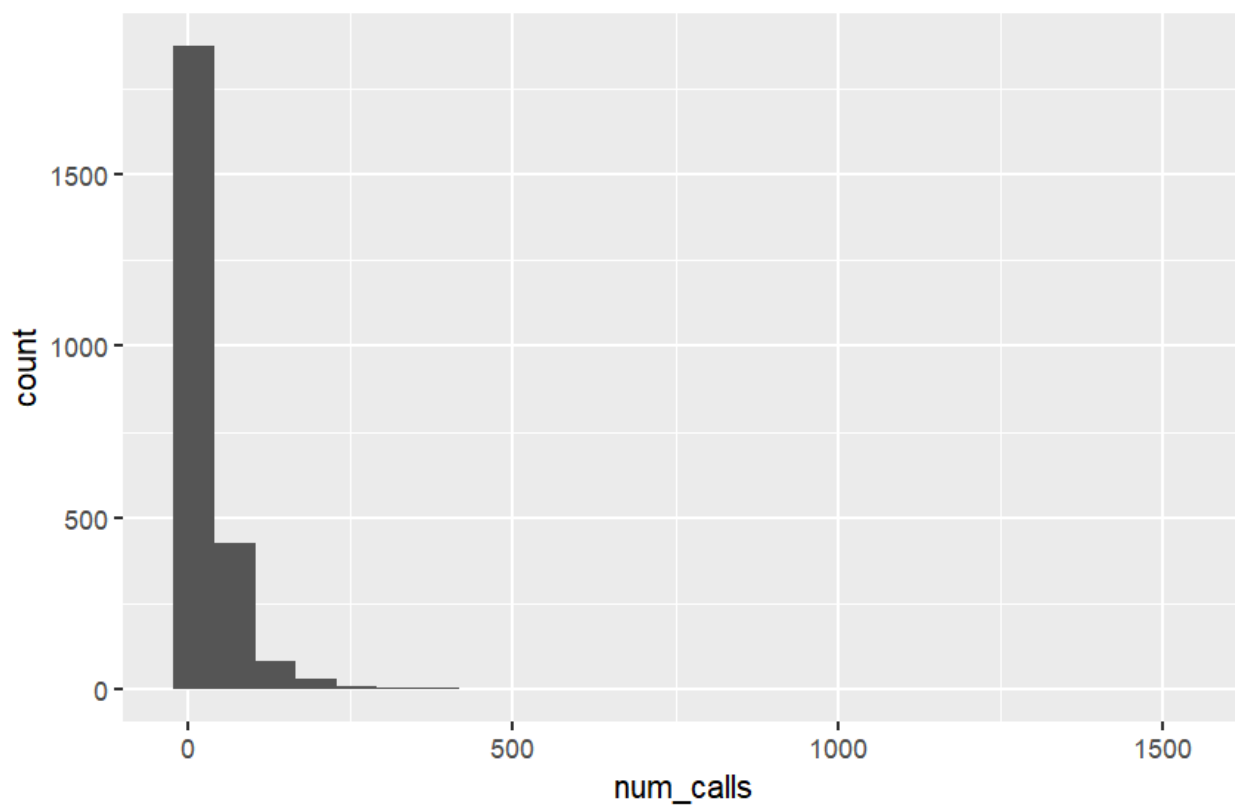
Hypothesis Tests

Because the means of Queens, Brooklyn and Manhattan are close enough to each other, there is a reason to conduct hypothesis tests. μ_1, μ_2, μ_3 will represent the means for each of the 3 Boroughs. I will conduct 3 hypothesis tests between each of them to see if there is any statistically significant evidence the average number of noise complaints per club differs. Alpha will be set at 0.05, which is the probability of a type I error that we can accept. Type I being the probability of saying there is a statistically significant difference between the 3 boroughs, when there is in fact not. H_0 will be $\mu_1 = \mu_2, \mu_1 = \mu_3$ and $\mu_2 = \mu_3$. $H_a = \mu_1 \neq \mu_2, \mu_2 \neq \mu_3$ and $\mu_1 \neq \mu_3$. The respective p-values were 0.7162 between Queens and Brooklyn, 0.5115 between Queens and Manhattan and 0.815 between Brooklyn and Manhattan. All of these P-values were significantly higher than alpha, meaning that we fail to reject the null hypothesis. Hence, we can

say that there was between a 51 and 82 percent chance of observing a result as or more extreme than one we observed if the null hypothesis were true.

Conclusion

The simulation results indicated meaningful variation in both the estimated means and the variability of those estimates across boroughs. While Brooklyn and Manhattan exhibited higher estimated average complaint levels, Bronx and Staten Island showed lower averages and substantially lower estimator variability. However, despite differences in variability and point estimates, the inferential procedures did not provide strong statistical evidence of differences in mean complaint levels across Brooklyn, Queens or Manhattan. The constructed confidence intervals showed substantial overlap, and the hypothesis tests failed to reject the null hypothesis in all comparisons. My final conclusion is that Brooklyn, Queens and Manhattan have on average clubs that generate more noise complaints compared to Staten Island and the Bronx. However, there is not enough evidence to conclude that a single Borough on average has the loudest clubs in New York.



	Borough	num	avg
	<i><chr></i>	<i><int></i>	<i><dbl></i>
1	BRONX	141	26.0
2	BROOKLYN	736	37.8
3	MANHATTAN	<u>1076</u>	37.1
4	QUEENS	418	39.2
5	STATEN ISLAND	64	35.2

	 Borough 	num 	avg 	std 	lower 	upper 
1	BRONX	141	26.04255	27.62113	21.483444	30.60166
2	BROOKLYN	736	37.84239	70.58606	32.742885	42.94190
3	MANHATTAN	1076	37.10595	57.87917	33.647635	40.56426
4	QUEENS	418	39.18900	53.87529	34.024242	44.35375
5	STATEN ISLAND	64	35.25000	31.65839	27.493836	43.00616

SOURCES

[2016 Parties in New York](#)

[NYC Open Data -](#)